

Syllabus:- Circle

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The equation of the smallest circle passing through the intersection of $x^2 + y^2 - 2x - 4y - 4 = 0$ and the line $x + y - 4 = 0$ is –
 - $x^2 + y^2 - 3x - 5y - 8 = 0$
 - $x^2 + y^2 - x - 3y = 0$
 - $x^2 + y^2 - 3x - 5y = 0$
 - $x^2 + y^2 - x - 3y - 8 = 0$
- Equation of a circle of radius 2 and touching the circles $x^2 + y^2 - 4|x| = 0$ is
 - $x^2 + y^2 \pm 4\sqrt{3}x + 8 = 0$
 - $x^2 + y^2 \pm 4\sqrt{3}y + 8 = 0$
 - $x^2 + y^2 \pm 4\sqrt{3}y - 8 = 0$
 - $x^2 + y^2 \pm 2\sqrt{3}y + 2 = 0$
- For the circle $x^2 + y^2 - 2x - 6y + 1 = 0$ the chord of minimum length and passing through (1, 2) is of length-
 - $2\sqrt{2}$
 - $4\sqrt{2}$
 - $6\sqrt{2}$
 - $8\sqrt{2}$
- If the chord $y = mx + 1$ of circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle, then the value of m is
 - $2 \pm \sqrt{2}$
 - $-2 \pm \sqrt{2}$
 - $-1 \pm \sqrt{2}$
 - none of these
- The equation of circle with origin as the centre and passing through vertices of an equilateral triangle whose median is of length 3a, is
 - $x^2 + y^2 = 9a^2$
 - $x^2 + y^2 = 16a^2$
 - $x^2 + y^2 = 4a^2$
 - $x^2 + y^2 = a^2$
- A variable circle passes through fixed point A(p,q) and touches x-axis. The locus of the other end of the diameter through A is
 - $(x - p)^2 = 4qy$
 - $(x - q)^2 = 4py$
 - $(y - p)^2 = 4qx$
 - $(y - q)^2 = 4px$
- The intercept on the line $y = x$ by circle $x^2 + y^2 - 2x = 0$ is AB. Equation of circle on AB as a diameter, is
 - $x^2 + y^2 - x - y = 0$
 - $x^2 + y^2 - x + y = 0$
 - $x^2 + y^2 + x + y = 0$
 - $x^2 + y^2 + x - y = 0$
- The set of points P(x, y) such that their distance from (3, 0) is $\sqrt{2}$ times their distance from (0, 2) form a circle-
 - whose radius is greater than 5 units
 - whose radius is equal to 5 units
 - whose radius is less than 5 units
 - None of these
- Triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have coordinates (3, 4) and (-4, 3) respectively then $\angle QPR =$
 - $\frac{\pi}{2}$
 - $\frac{\pi}{3}$
 - $\frac{\pi}{4}$
 - $\frac{\pi}{6}$

10. The equation of the circle, which is the mirror image of the circle, $x^2 + y^2 - 2x = 0$, in the line, $y = 3 - x$ is
 (1) $x^2 + y^2 - 6x - 4y + 12 = 0$ (2) $x^2 + y^2 - 6x - 8y + 24 = 0$
 (3) $x^2 + y^2 - 8x - 6y + 24 = 0$ (4) $x^2 + y^2 - 4x - 6y + 12 = 0$
11. Consider
 $L_1: 3x + y + \alpha - 2 = 0$;
 $L_2: 3x + y - \alpha + 3 = 0$, where α is a positive real number, and
 $C: x^2 + y^2 - 2x + 4y - 4 = 0$.
Statement 1 : If line L_1 is a chord of the circle C , then the line L_2 is not always a diameter of the circle C .
Statement 2 : If line L_1 is a diameter of the circle C , then the line L_2 is not a chord of the circle C .
 Then,
 (1) Statement - 1 is true ; Statement - 2 is false.
 (2) Statement - 2 is true and statement-1 is false
 (3) both the statements are true.
 (4) both the statements are false.
12. Consider the circle $S: x^2 + y^2 - 1 = 0$ and the line $L: y - m(x + 2) = 0$ where m is a parameter. If L intersects S at P and Q , then the locus of middle point of PQ is
 (1) $(x+1)^2 + y^2 = 1$ (2) $x^2 + (y-1)^2 = 1$ (3) $(x-1)^2 + y^2 = 1$ (4) $x^2 + (y+1)^2 = 1$
13. If S is a circle with centre C which cuts the pair of straight lines $x^2 + xy - 3x - 3y = 0$ orthogonally and also cuts the circle $S_1: x^2 + y^2 - 2x - 4y + 1 = 0$ at A and B such that $\angle CAC_1 = \frac{\pi}{2}$, where C_1 is the centre of the circle S_1 , then find the radius of the circle S .
 (1) 3 (2) 5 (3) 7 (4) 9
14. A variable circle C touches the line $y = 0$ and passes through point $(0, 1)$. The locus of centre of circle C is a conic S . Equation of director circle of conic S is
 (1) $x = 0$ (2) $y = 0$ (3) $x = 2$ (4) $y = \frac{1}{2}$
15. If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - d = 0$ ($c, d > 0$), then maximum value of cd is
 (1) 25 (2) 125 (3) 425 (4) 625
16. What will be the equation of circle whose centre is $(3, -1)$ and which intercept chord of 6 unit length on straight line $2x - 5y + 18 = 0$
 (1) $x^2 + y^2 - 6x + 2y - 28 = 0$ (2) $x^2 + y^2 - 6x + 2y + 1 = 0$
 (3) $x^2 + y^2 - 6x + 2y - 1 = 0$ (4) $x^2 + y^2 - 6x + 2y - 12 = 0$
17. The line $lx + my = 1$ intersects the circle $x^2 + y^2 = a^2$ at points A, B , if AB subtends 45° at the origin, then $a^2 (l^2 + m^2) =$
 (1) $4 \pm 2\sqrt{2}$ (2) $4 \pm 2\sqrt{6}$ (3) $2\sqrt{6}$ (4) $4 - \sqrt{6}$
18. Equation of tangents to the $x^2 + y^2 - 2x - 4y + 1 = 0$ which are parallel to the x -axis are
 (1) $y = 0$; $y = 2$ (2) $y = 0$; $y = 4$ (3) $y = 2$; $y = 4$ (4) $y = 2$; $y = -2$
19. If the line $x + 2by + 7 = 0$ is a diameter of the circle $x^2 + y^2 - 6x + 2y = 0$, then $b =$
 (1) 5 (2) 4 (3) 2 (4) 3
20. If two circles of radius r with centre $(2, 3)$ and $(5, 6)$ cut orthogonally then ' r ' is equal to
 (1) 5 (2) 4 (3) 2 (4) 3

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. If the line segment joining $P(1, 3)$ and $Q(5, 7)$ subtends a right angle at $R(x, y)$ and area of $\Delta PQR = 2$ then find the maximum number of such points R in the xy -plane
22. If circles $x^2 + y^2 + 2gx + 2y + 2 = 0$ and $x^2 + y^2 + 2x + 2fy + 4 = 0$ intersect in two distinct points P and Q , and equation of line PQ is $x + y + 1 = 0$ then find the value of $g + f$.
23. Find the number of integer values of k for which the equation $x^2 + y^2 + (k - 1)x - ky + 5 = 0$ represents a circle whose radius cannot exceed 3.
24. If the tangent at point P on circle $x^2 + y^2 + 6x + 6y = 2$ meets straight line $5x - 2y + 6 = 0$ at point Q on y -axis then find length PQ
25. Find the number of integer values of k for which the equation $x^2 + y^2 + (k - 1)x - ky + 5 = 0$ represents a circle whose radius cannot exceed 3.

ANSWER KEY_DPP-08

1.	(3)	2.	(2)	3.	(2)	4.	(4)	5.	(3)	6.	(1)	7.	(1)
8.	(1)	9.	(3)	10.	(1)	11.	(1)	12.	(1)	13.	(2)	14.	(2)
15.	(4)	16.	(1)	17.	(1)	18.	(2)	19.	(1)	20.	(4)	21.	(4)
22.	(2)	23.	(4)	24.	(5)	25.	(4)						